

G-02-epsilon0-import v1: audit ESS-2003 -i local concentration C_G

ZFL Lab

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EXTERNAL-DEPENDENCY: ESS-2003 (Escauriaza-Seregin-Sverak) INPUT: C_{L^3} PROVEN: $T_* < \infty \implies \limsup \|u\|_3 = +\infty$ TARGET: C_G : $\exists t_n \uparrow T_*, x_n, r_n \downarrow 0 : \|u(t_n)\|_{L^3(B(x_n, r_n))} \geq \epsilon_0$

AUDIT:

1. What ESS-2003 actually proves: If u suitable weak in $R^3 \times (-1, 0)$ with $\|u\|_{L_t^\infty L_x^3} \leq M < \infty$, then regular up to 0. Contrapositive gives C_{L^3} .

2. Does ESS imply C_G ? ESS gives global criterion, not local ball lower bound with $r_n \rightarrow 0$. To get C_G need concentration-compactness: rescale near blow-up, extract profile. If no concentration, scaling limit would be bounded L^3 regular, contradicting blow-up. Requires backward uniqueness + profile decomposition, not stated as C_G in ESS.

Thus:

ESS-2003 does NOT directly imply C_G as defined

3. What additional bridge needed? (i) Local ϵ -regularity: $\|u\|_{L_t^\infty L^3(B_1)} \leq \epsilon_*$ implies regularity. This is Kozono-Taniuchi / ESS local version, distinct from global ESS. (ii) $r_n \downarrow 0$ guarantee requires parabolic scaling $r_n \sim \sqrt{T_* - t_n}$.

4. Is ϵ_0 universal? Universal only if local version holds, not from global ESS constant alone.

5. Extra hypotheses: suitable weak, finite energy, pressure in $L_{loc}^{3/2}$.

STATUS:

C_G^{cond} NOT DERIVED from ESS-2003 alone

C_G remains OPEN, requires local epsilon-regularity + concentration

ESS $\implies C_{L^3}$ PROVEN, but ESS $\not\implies C_G$ directly

Hence $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ untouched.

Next: G-03 for local version.

FORBIDDEN respected: No claim “ESS implies C_G ” without explicit bridge.